Comparative Analysis of Selected Research Papers

SL No.	References	AI/ML Technique	Healthcare Application	Dataset Used	Key Contributions	Publisher
1	Yun et al. (2023)	Federated Learning, LSTM, BERT	Medical NLP / Clinical Text Analysis	Clinical notes and medical records	Integrates FL with NVFlare framework for privacy-preserving medical NLP; demonstrates BERT pretraining for context understanding in clinical data	IEEE
2	Poulain et al. (2023)	Federated Learning, Adversarial Debiasing	Fairness in AI / EHR	Electronic Health Records	Proposes comprehensive FL approach with adversarial debiasing and fair aggregation; explicitly mitigates bias as part of optimization process	ACM
3	Georgoutsos et al. (2023)	Federated Learning (FedAvg, FedProx), RNN	ICU Mortality Prediction	Multi-center ICU benchmark	FL models outperform local ML models on non-IID distributed data; demonstrates FL as promising solution for ICU mortality prediction	ACM
4	Lee et al. (2024))	Federated Learning, CNN	Pediatric Brain Tumor Classification & Segmentation	Multi-center pediatric brain tumor data (19 sites)	FL-PedBrain platform enables training across 19 international sites; <1.5% decrease in classification vs centralized; 20-30% segmentation boost on external sites	Nature Communications
5	Nafis et al. (2023)	Federated Learning, CNN, Ensemble Learning	Lung Disease Detection	Chest X-ray images	Compares sequential and ensemble models for FL and centralized approaches; demonstrates FL effectiveness for lung cancer and tuberculosis detection	ACM
6	Chung & Lee (2024)	Federated Influencer Learning (FIL)	Secure Collaborative Learning / Medical Imaging	X-ray, MRI, PET medical images	Proposes FIL as secure and efficient collaborative learning paradigm; eliminates model-parameter transactions and central servers;	Nature (Scientific Reports)
7	Thakur et al. (2024)	Federated Learning, Knowledge Abstraction	Heterogeneous EHR / Privacy-Preserving Analytics	Three healthcare datasets	Addresses data view heterogeneity without manual alignment or information loss; maps raw inputs to unified semantic space	Nature (npj Digital Medicine)
8	Chanda et al. (2024)	Explainable AI (XAI), Deep Neural Networks	Melanoma Diagnosis	Skin lesion images	Provides precise, domain-specific XAI explanations alongside differential diagnoses; increases dermatologist confidence and trust significantly	Nature Communications
9	Abbas et al. (2024)	Adaptive Federated Learning, Deep Learning	Multidisciplinary Cancer Classification	Cancer disease datasets	Self-adaptive FL framework achieves 90.0% accuracy; outperforms conventional FL (87.30%) for multidisciplinary cancer prediction	Springer (Scientific Reports)
10	Suttaket & Kok (2024)	Rational Multi-Layer Perceptrons (RMLP)	Interpretable Predictive Modeling / EHR	Real-world Electronic Health Records	Pioneers RMLP model from weighted finite state automata; provides better interpretability by linking relevant inputs into distinct sequences	ACM
11	Yang et al. (2024)	Federated Learning, GAN, Knowledge Distillation	Medical Imaging Segmentation	COVID-19 and prostate MRI datasets	Proposes MixFedGAN framework with dynamic aggregation and knowledge distillation; addresses non-IID data and limited labeling challenges	MDPI (Electronics)

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12	Nazir & Kaleem (2024)	Federated Learning, Deep Neural Networks	Medical Image Analysis	Medical imaging datasets	Reviews FL applications in medical image analysis with DNNs; highlights security concerns and future research directions	MDPI (Diagnostics)
13	Tawfik et al. (2025)	Homomorphic Encryption, Secure Multiparty Computation, Blockchain	Privacy-Preserving Collaborative Analysis	Healthcare records on blockchain	Integrate secret-sharing, SMPC, and homomorphic encryption within blockchain; enables secure collaborative statistical analysis on health records	Springer (Cluster Computing)
14	Bai et al. (2024)	Explainable AI, Feature Grouping, Model Comparison	Breast Cancer Diagnosis	Mammogram and biomarker data	Systematically evaluates multiple XAI methods for breast cancer diagnosis; enhances interpretability through feature grouping and model comparison	IEEE
15	Xu et al. (2023)	Heterogeneous FL, Differential Privacy, Voting Mechanism	Sensitive Healthcare Data Privacy / Disease Prediction	Diabetes and inpatient mortality data	Proposes AAFV framework with differential privacy and abstention voting for heterogeneous local model collaboration with privacy protection	IEEE
16	Peng et al. (2024)	Federated Learning, Biomedical NLP	Biomedical Information Extraction	8 biomedical NLP corpora	Evaluates FL on 2 biomedical NLP tasks using 6 language models; demonstrates feasibility of FL for privacy-preserving clinical NLP	Springer (npj Digital Medicine)
17	Pan et al. (2024)	Federated Learning, CNN	Pediatric Pneumonia Detection	Chest X-ray images	Achieves high accuracy for pediatric pneumonia classification using FL; addresses data heterogeneity across institutions	Springer (Scientific Reports)
18	Kim et al. (2023)	Explainable AI, Deep Learning	Clinical Decision Support Systems	Multiple clinical datasets	Proposes new XAI-based CDSS framework; provides resources, datasets, and models for multi-disease decision support	MDPI (Applied Sciences)
19	Wang et al. (2025)	Server-Rotating Federated Learning, Differential Privacy	Cross-Vendor Diagnostic Imaging	Multi-vendor medical imaging data	Proposes SRFLM with rotational server-communication and dynamic server-election; provides strong privacy guarantees using differential privacy	Nature (Communications Engineering)
20	Haripriya et al. (2025)	Federated Learning, Transfer Learning, ResNet, VGG16	Medical Image Classification	TB chest X-rays, brain tumor MRI, diabetic retinopathy images	Combines FL with transfer learning for privacy-enhanced medical image classification; scalable and privacy-preserving framework	Springer (Journal of Big Data)

Summary of Comparative Analysis:

The selected studies illustrate that Federated Learning is a feasible approach for healthcare AI that protects privacy, allowing safe cooperation amongst dispersed medical facilities while preserving competitive model performance. The literature highlights developments in explainability, scalability, privacy, security, and fairness, offering a thorough basis for the suggested study on lung disease diagnosis.